Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Chen, Lea Der Ph.D.		
Project Title:	Reflight of ELF (Enclosed Laminar Flames) Investigation		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ldchen@engineering.uiowa.edu	**Fax:**	FY 319-384-0576
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-5674
Organization Name:	The University of Iowa		
PI Address 1:	3131 SC, Deparment of Mechanical & Industrial Engineering		
PI Address 2:	The University of Iowa		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242	**Congressional District:**	2
Comments:	NOTE (June 2016): Dr. Chen is currently at Texas A&M Corpus Christi		
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/1999	**End Date:**	01/03/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stocker, D.P. (NASA Glenn Research Center) Brooker, J.E. (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3 666		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	A joint experimental and computational approach is adopted. Data obtained from ELF Glovebox Investigation were used to evaluate the effects of fuel Schmidt number on the transition from the lifted flame to the blow-out condition. Comparison tests were conducted in normal gravity to allow isolation and identification of the influence of buoyancy. The results were used to map the liftoff and blowout stability limits as a function of the fuel and air velocities for the two buoyancy conditions. A methane and nitrogen mixture (50% on volume basis) was used as the fuel. The experimental results verified the hypothesis that substantially greater velocities are required to destabilize the flame in microgravity. The increase in air velocity required to induce liftoff in microgravity (compared to normal gravity) was roughly equal to the increase required to induce blowout in microgravity. Furthermore, the air velocity increase was relatively independent of the fuel flow, except at low fuel flows. At high fuel flows, the microgravity flames tend to immediately blow out after lift off, in agreement with the free-jet theory in the literature for fuels with a Schmidt number less than 0.7. However, stable lifted flames were achieved within a coflow duct in microgravity at low fuel flows, and in normal gravity. A numerical code with adaptive mesh refinement was developed and used to solve the transport equations, including the finite rate chemistry based on a four-step reduced mechanism. The calculated blowout conditions were in quantitative agreement with ELF results for fuel jet velocity below 0.2 m/s, but only in qualitative agreement for higher fuel jet velocities. The overall objective of the program is to study the burner stability of enclosed laminar flames in microgravity environments. The specific objectives are (a) to evaluate the effects of fuel Schmidt number on the transition from lifted flames to blow-out conditions, (b) to study the hysteresis effects observed in the reattachment of lifted flames, (c) to gain physical insight into the transition from lifted to blow-out conditions, and (d) to validate numerical simulation. Ground-based experiments are conducted to support the space experiments The current project studies the flame stabilization in coflow environments. The findings will contribute to the fundamentals of combustion theory. The current invetsigation is also of practical significance as practical combustors often have jet flames in flow environments, for example, co-flow inside gas turbine engine combustors and in industrial furnaces. Understanding of the flame stabilization at the fundamental level is key to safety considerations in the design of new combustors. The verified numerical model for the prediction of the flame stabilization can be applied as a design tool for complex engineering systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This invetsigation provides additional insight of the structure of the edge flame from stabilized to (near) blowout conditions. Specifically, the ratio of length scales versus the ratio of time scales was extracted from the numerical results, and the ratio of length scales versus Damkohler number was used to characherize the flame regimes. The adpative mesh refinement flame code will also be used to study the deNOx process in catalytic converters.
Task Progress:	A numerical code with adaptive mesh refinement was developed and used to solve the transport equations, including the finite rate chemistry based on a four-step reduced mechanism. The calculation based on a four-step reduced mechanism was be able to predict the blowout conditions, and in quantitative agreement with ELF results in microgravity at low fuel jet velocities, e.g., 0.2 m/s, but in qualitative agreement for higher fuel jet velocities. The numerical calculation also shows that the blowout condition is most sensitive to the heat loss level and oxygen content in ambient air. The calculated blowout velocity based on the four-step reduced mechanism is in agreement with that based on the full starting mechanism for the conditions examined. The calculated reaction zone structures are qualitatively similar, but with noted differences... The flame leading edge exhibits significantly different structure, whereas similar structures of the trailing diffusion flames are calculated. Error of the radical diffusive fluxes at the flame leading edge calculated based on the reduced mechanism was responsible for the differences at the leading edge. The four-step reduced mechanism also calculates a different evolution of the flame from a stabilized condition to the blowout although it was able to calculate the blowout air velocity with sufficient accuracy. Comparison of the results shows that the four-step reduced mechanism can properly calculate the major species mass fraction, temperature, velocity, and reaction rates as compared to the results based on the full starting mechanism. It is noted that the computational time requirement of the full starting mechanism is about 3 to 4 times of that based on the four-step reduced mechanism, which is of the order of month for calculation of a single blowout condition. The excessive computational effort required by the full starting mechanism is due to the fact that the number of transport equations is increased, and that a much smaller time step is required. The four-step reduced mechanism seems can be used to calculate the blowout conditions of methane fueled laminar diffusion flames if the blowout out velocity is information of primary interest.
Bibliography Type:	Description: (Last Updated: 06/13/2007)
Articles in Peer-reviewed Journals	Venuturumilli, R.; Chen, L.-D. "Numerical Simulation of Laminar Jet Diffusion Flames Using an Adaptive Grid" Numerical Heat Transfer B: Fundamentals, Vol. 46 , Aug-2004